

Protik Dey

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RESEARCH SUMMARY

I study how blind and low-vision (BLV) people interact with Generative AI (GenAI), combining human-computer interaction methods with technical prototyping to understand where these systems build trust and where they introduce new privacy and accessibility risks. My work pairs qualitative research like interviews and user studies with BLV participants with the design and evaluation of AI-based assistive prototypes, grounding findings in both lived experience and system behavior.

RESEARCH AREA

Human-Computer Interaction, Trustworthy AI, Explainable AI, Usable Privacy, Accessibility

EDUCATION

Ph.D. in Computer Science

Aug 2024 – Present

University of Texas at San Antonio, Texas, USA

Advisor: [Dr. Taslima Akter](#) | [APEX Lab](#)

B.Sc. (Engg.) in Computer Science and Engineering

Feb 2016 – Mar 2021

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh

Thesis: *Malicious Code Detection and Analysis*, advised by Dr. Muhammad Masroor Ali

RESEARCH EXPERIENCE

Graduate Research Assistant II

Jan 2026 – Present

University of Texas at San Antonio, Texas, USA

- Lead research on accessible, trustworthy, and privacy-aware generative AI for blind and low-vision (BLV) users.
- Design and conduct user studies with BLV participants; build research prototypes to evaluate accessible AI interactions, from study design and interviews through qualitative analysis and publication.
- Mentor undergraduate students on research methods and academic writing.

Fake Bengali Facebook Post Detection on COVID-19

2022

Research Project

- Built a benchmark dataset of Bengali-language Facebook posts on COVID-19, collected via CrowdTangle.
- Trained transformer-based models on the dataset and reported on the correlation between misinformation and audience engagement, one of the first empirical studies of Bengali-language misinformation during the pandemic.

Malicious Code Detection and Analysis

2020 – 2021

B.Sc. Thesis

- Extended CodeOntology to express the semantic structure of Java source code as an ontology, encoding characteristic malware signatures (e.g., repeated connection calls underlying a DDoS attack).

- Identified a known malware from source code alone and produced human-interpretable explanations for flagged code; the approach remained robust against obfuscation techniques that scatter malicious calls across intermediate methods.

Performance Analysis of Highly Dense Terahertz Communication Networks

2019

Research Project

- Simulated 5G and beyond-5G network communication using NS-3.
- Analyzed throughput and scaling behavior of highly dense terahertz communication networks.

PUBLICATIONS

Peer-Reviewed Papers

1. Syed Zami-Ul-Haque Navid, **Protik Dey**, Shamiul Hasan, and Muhammad Masroor Ali, “Static Detection of Malicious Code in Programs Using Semantic Techniques”, In the Proceedings of the 11th International Conference on Electrical and Computer Engineering (**ICECE 2020**).

Workshop Papers

1. **Protik Dey**, Mohd. Saifuzzaman, and Taslima Akter, “More Than Just Access: Generative AI as Communication Intermediary for Blind and Low-Vision Users”, **ACM CSCW 2026 Workshop on Broader Impacts of GenAI in Communication: Building Agendas for Research, Design, and Policymaking**.
2. Abu Noman Md Sakib, **Protik Dey**, Zijie Zhang, and Taslima Akter, “Explainable AI for Blind and Low-Vision Users: Navigating Trust, Modality, and Interpretability in the Agentic Era”, ACM CHI Workshop on Human-Centered Explainable AI (**HCCXAI 2026**).

Preprints

1. Protik Bose Pranto, Syed Zami-Ul-Haque Navid, **Protik Dey**, Gias Uddin, and Anindya Iqbal, “Are You Misinformed? A Study of COVID-Related Fake News in Bengali on Facebook”, **arXiv preprint, 2022**.

Posters

1. **Protik Dey**, Mohd. Saifuzzaman, and Taslima Akter, “From TRAP to PrivacyGuard: On-Device, Pre-Disclosure Privacy Protection for Blind and Low-Vision Users”, **Texas Security Workshop, 2026**.
2. **Protik Dey**, Mohd. Saifuzzaman, and Taslima Akter, “Trustworthy GenAI Use Among Blind and Low-Vision People”, **NSF AI Spring School, 2026**.
3. Nirob Arefin, **Protik Dey**, and A.B.M. Alim Al Islam, “Throughput Analysis of Highly Dense Terahertz Communication Network”, 6th International Conference on Networking, Systems and Security (**NSysS 2019**).

TEACHING EXPERIENCE

Graduate Teaching Assistant

Aug 2024 – Dec 2025

University of Texas at San Antonio, Texas, USA

- Teaching assistant for undergraduate Operating Systems and Systems Programming.
- Graded assignments, projects, and exams; supported students through office hours on OS concepts and low-level C programming.

INDUSTRY EXPERIENCE

Software Engineer II, Kona Software Lab Limited

Jan 2023 – Jul 2024

Integrated Platform Team, Dhaka, Bangladesh

- Worked in the Integrated Platform team building global payment solutions.
- Designed and implemented validation and authorization for financial transactions, including recharge and payment flows.
- *Tech Stack: Spring Boot, Redis, RabbitMQ, MariaDB*

Graduate Engineer, Kona Software Lab Limited

Aug 2021 – Dec 2022

Digital Financial System Team, Dhaka, Bangladesh

- Full-stack developer on the Digital Financial System team, delivering financial transaction solutions for Nagad, Bangladesh's fastest-growing mobile financial system, including tokenized payments, KYC, and bank/card transfers.
- Helped stand up the back-office portal for the KONAPLATE project using Angular.
- *Tech Stack: Spring Boot, Angular, Redis, RabbitMQ, Strapi, Oracle*

PROFESSIONAL SERVICE

External Reviewer

- CHI 2026
- ASSETS 2025
- CSCW 2025, 2026

Volunteering

- Volunteer at a BLV community event — [Vibrant Works](#), San Antonio April 2026

MENTORING

- **Blessy Kalluri**, Undergraduate Student, University of Texas at San Antonio Summer 2026
- **Abhinaya Guduru**, Undergraduate Student, University of Texas at San Antonio Spring – Fall 2025
- **Keylee Chambers**, Undergraduate Student, University of Texas at San Antonio Spring – Summer 2025

SKILLS

- **HCI & Research Methods:** User studies, thematic analysis, usability testing, interactive prototyping, accessible & inclusive design
- **Programming & Data:** Python, C, C++, Java, JavaScript, TypeScript, Bash, NumPy, Pandas, Scikit-learn, Matplotlib
- **Engineering Stack:** Spring Boot, Django, Angular, Redis, RabbitMQ, Kafka, Hibernate, Oracle, MySQL, PostgreSQL, MongoDB, Git